



DR ESLAM AHMED
eliteigcse.com

Venn and Tree Diagrams

Venn diagram: fill from the middle outwards. Tree diagram: multiply along the branches, add between the paths.

THE TOPIC ON ONE MAP

Venn and Tree Diagrams

01 Two-circle Venn

$n, U, '$

fill from the middle

"given that"

02 Three-circle Venn

all three first

pairs, then singles

03 Trees: independent

$\times$ along, $+$ between

04 Trees: dependent

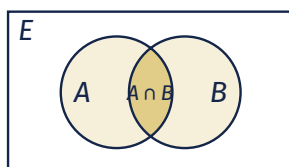
no replacement

"if ... then"

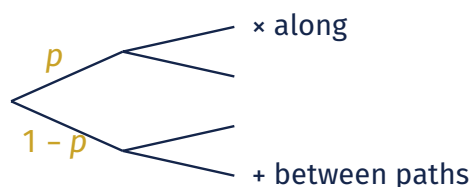
05 Longer trees

three stages

TWO DIAGRAMS Which one? Groups that overlap $\rightarrow$ Venn. Events one after another $\rightarrow$ tree.



overlapping groups: VENN



one event after another: TREE

01 Two-Circle Venn Diagrams

Four regions: A only, both, B only, outside. Always fill **both** first (the overlap), then A only = $n(A) - \text{both}$, then B only, then outside = total - everything in the circles.

WHEN YOU SEE → YOU DO

$A \cap B$	→ in both — the overlap
$A \cup B$	→ in A or B or both — everything inside the circles
A'	→ not in A — including outside
$n(A \cup B)$ given	→ overlap = $n(A) + n(B) - n(A \cup B)$
“chosen from those who ...”	→ the bottom of the fraction is only that group

ZONE $A \cap B$

both

the lens

ZONE $A \cup B$

either

both circles

ZONE A'

not A

outside circle A

ZONE $A \cap B'$

A only

in A, not in B

WORKED MODEL • JANUARY 2021 P2HR

$n(E) = 80$, $n(A) = 62$, $n(B) = 30$, $n(A \cup B) = 68$. Find $P(A \cap B)$.

1 both = $62 + 30 - 68 = 24$

overlap first

2 A only 38, B only 6, outside $80 - 68 = 12$

fill outwards

ANSWER $P(A \cap B) = \frac{24}{80} = \frac{3}{10}$

× **TRAP** Putting $n(A) = 62$ in the “A only” region. The circle’s number includes the overlap.

IDEA 01

Practise: Two-Circle Venn Diagrams

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Fill the regionsCH 6 • TOPIC 5 **3 marks**

$n(E) = 40$, $n(A) = 22$, $n(B) = 15$, $n(A \cap B) = 6$. Find the number in each region and $P(A \cup B)$.

SELECTED PRACTICE

Q2

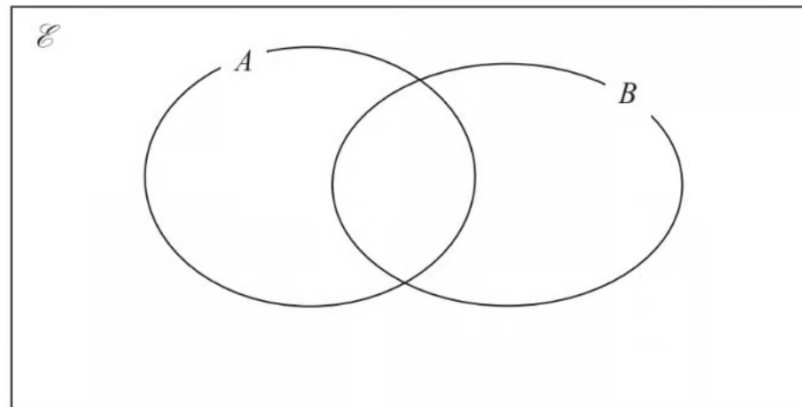
PRACTICE

Practice • Easy

CH 6 • TOPIC 5

5 marks

- $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
1 (a) $A = \{1, 5, 6, 8, 9\}$
 $B = \{2, 6, 9\}$



Complete the Venn diagram to represent this information.

(3 marks)

- (b) A number is chosen at random from the universal set $\mathcal{E}$.

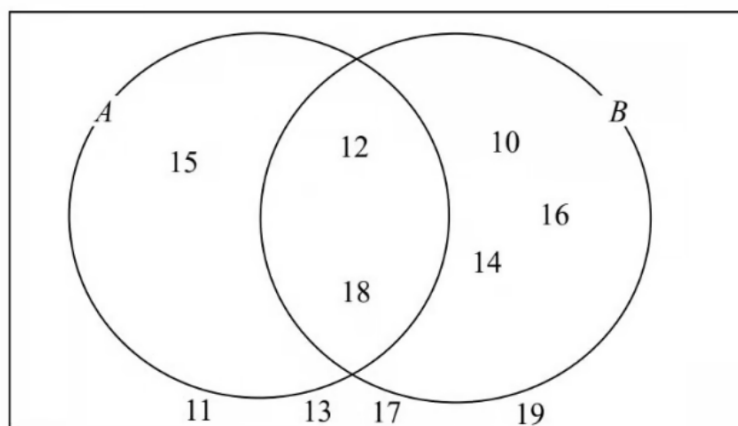
Find the probability that the number is in the set $A \cap B$.

(2 marks)

Q3**PRACTICE**

Practice • Medium

CH 6 • TOPIC 5 4 marks

2 (a) Here is a Venn diagram.

Write down the numbers that are in set

i) $A \cup B$

[1]

ii) $A \cap B$

[1]

(2 marks)**(b)** One of the numbers in the diagram is chosen at random.Find the probability that the number is in set A' .**(2 marks)**

Q4**PRACTICE**

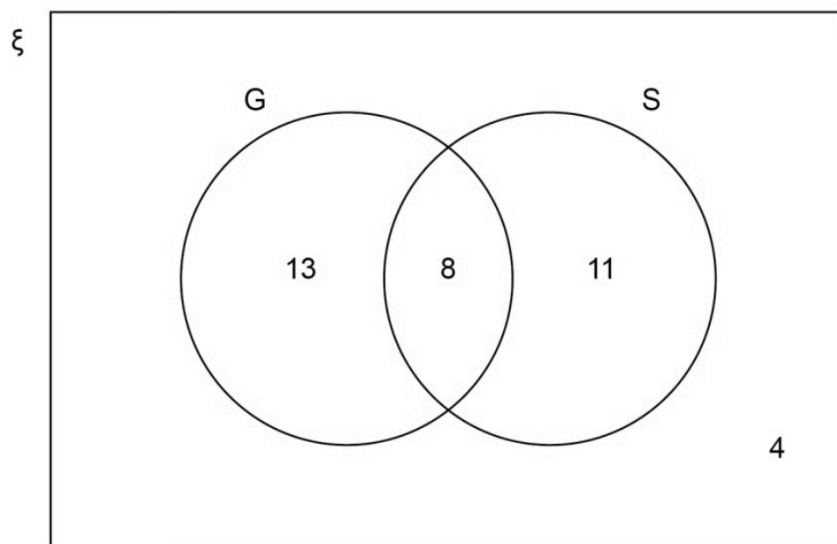
Practice • Hard

CH 6 • TOPIC 5 4 marks

19 (a) The Venn diagram shows information about some houses.

G = houses with a garage

S = houses with a shed



A house is chosen at random.

The house has a garage.

What is the probability that it has a shed?

(1 mark)

(b) The house does **not** have a garage.

What is the probability that it does **not** have a shed?

(1 mark)

(c) Show that $P(G \cap S)' > P(G \cup S')$

(2 marks)

Q5**PRACTICE**

Practice • Hard

CH 6 • TOPIC 5**5 marks**

26 In a group of 120 adults, 85 watch football, 78 play a sport and 20 do neither.

Find the probability that an adult chosen at random from those who watch football does not play a sport.

(5 marks)

PAST PAPERS

Q6

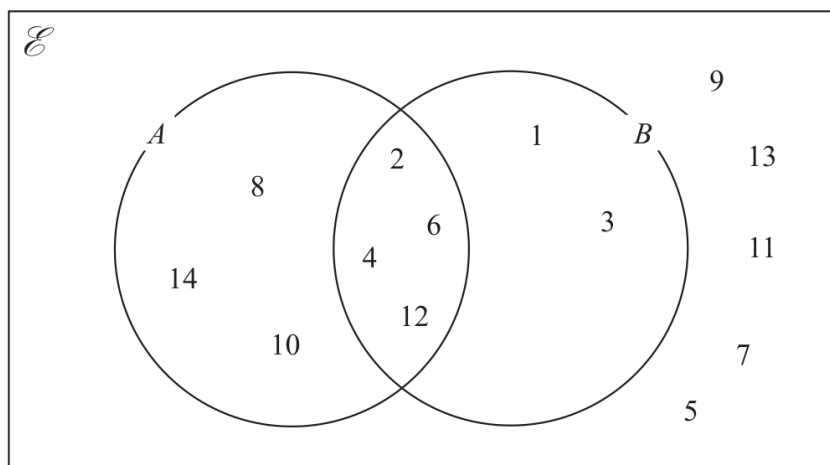
PAST PAPER

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q1

CH 6 • TOPIC 5 4 marks

You must write down all the stages in your working.

- 1 The numbers from 1 to 14 are shown in the Venn diagram.



- (a) List the members of the set $A \cap B$

.....

- (b) List the members of the set B'

.....

A number is picked at random from the numbers in the Venn diagram.

- (c) Find the probability that this number is in set A but is **not** in set B.

.....

(Total for Question 1 is 4 marks)

Q7**PAST PAPER**

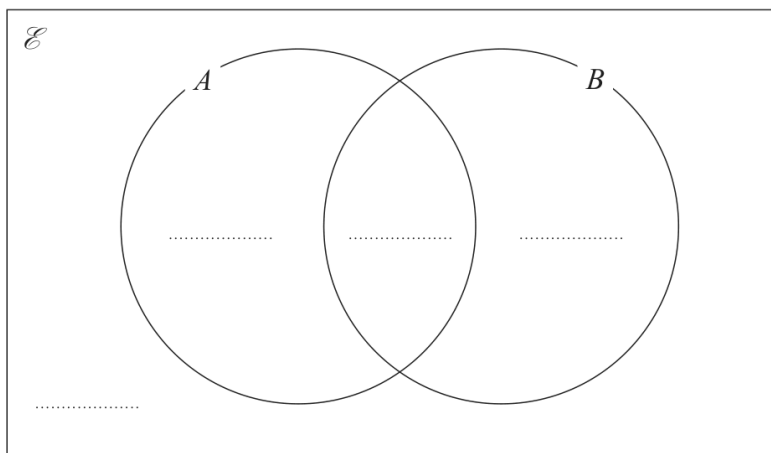
Edexcel 4MA1 • January 2021 • Paper 2HR • Q16

CH 6 • TOPIC 5 5 marks

16 Two events A and B are such that $n(A) = 62$ $n(B) = 30$ and $n(A \cup B) = 68$

Given that $n(\mathcal{E}) = 80$

(a) complete the Venn diagram to show the number of elements in each region.



An element is chosen at random from $\mathcal{E}$.

(b) Using the Venn diagram, find the probability that this element is in

(i) $A \cap B$

.....

(ii) $A \cup B'$

.....

(Total for Question 16 is 5 marks)

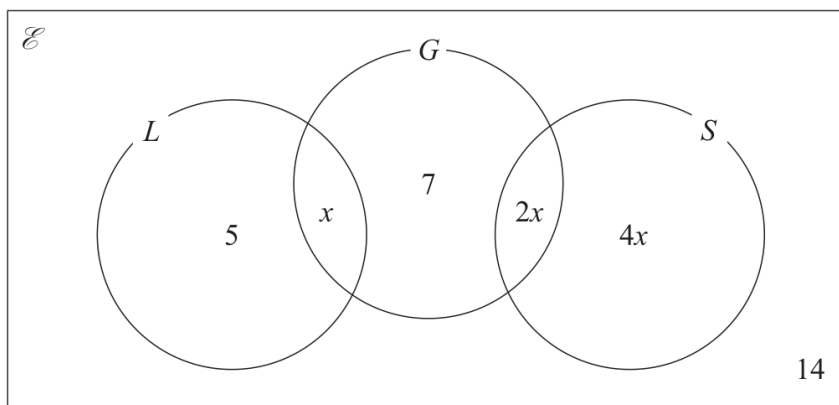
Q8**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q18

CH 6 • TOPIC 5 3 marks

- 18 Bianca asked 40 students which of the languages Latin (L), Greek (G) and Sanskrit (S) they study.

The Venn diagram gives some information about her results.
It shows the number of students in each subset.



One of these students is selected at random.
Work out the probability that this student studies Latin or Sanskrit.

(Total for Question 18 is 3 marks)

02 Three-Circle Venn Diagrams

Same rule, one more step: the centre (all three) first, then each pair **minus the centre**, then each single circle minus everything already inside it, then the outside.

WHEN YOU SEE → YOU DO

“6 have all three”

→ write 6 in the centre first

“11 have sheep and goats”

→ $11 - 6$ in the S-G only region

“31 have sheep”

→ S only = $31 -$ the three numbers already in S

an unknown x in the centre

→ write every region in x , then use one total

“one who has chickens ... also has goats”

→ $\frac{C \text{ and } G}{\text{all of } C}$

Centre

Pairs minus centre

Singles

Outside

WORKED MODEL • JUNE 2024 P2H

60 students. $K \cap P \cap E = 9$, $K \cap P = 17$, $P \cap E = 16$, $P = 28$. Given a student chose P , find $P(\text{also } K)$.

1 $KP \text{ only} = 17 - 9 = 8$, $PE \text{ only} = 16 - 9 = 7$

pairs minus centre

2 $P \text{ only} = 28 - 9 - 8 - 7 = 4$

single

ANSWER $P(K | P) = \frac{8+9}{28} = \frac{17}{28}$

IDEA 02

Practise: Three-Circle Venn Diagrams

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Middle first

CH 6 • TOPIC 5 3 marks

Three sets: all three = 4, $A \cap B = 10$. How many are in $A \cap B$ but not in C ?

SELECTED PRACTICE

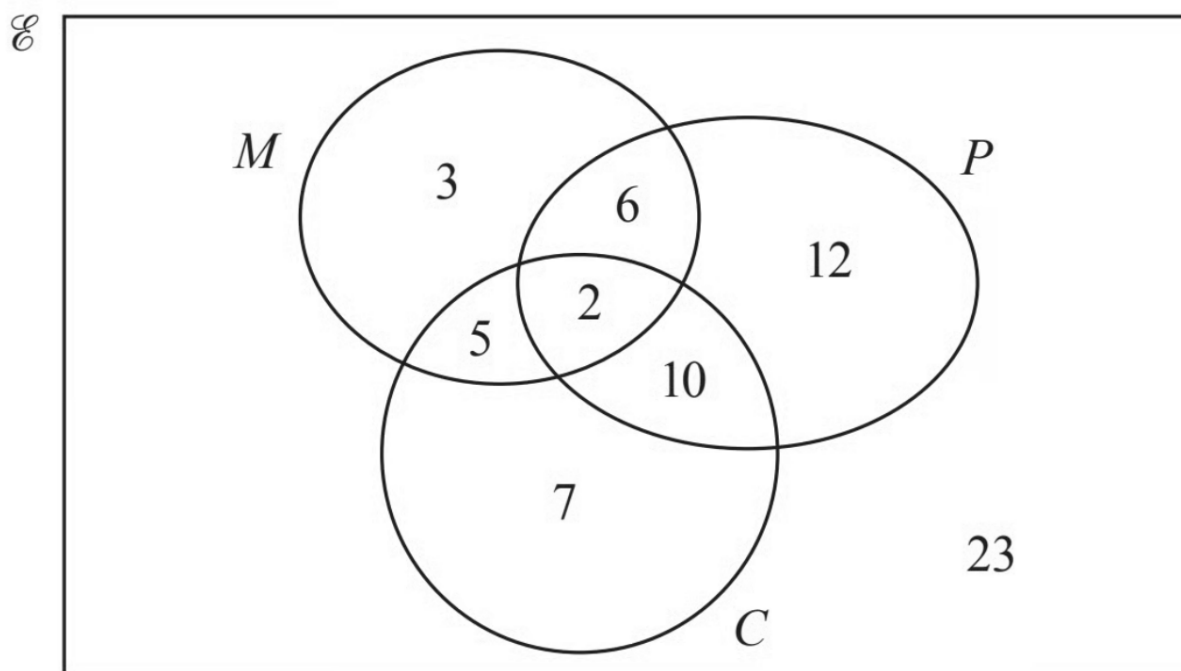
Q10

PRACTICE

Practice • Easy

CH 6 • TOPIC 5 2 marks

9 The Venn diagram below shows information about the number of gardeners who grow melons (M), potatoes (P) and carrots (C).



A gardener is chosen at random from the gardeners who grow melons.

Find the probability that this gardener does not grow carrots.

(2 marks)

Q11**PRACTICE**

Practice · Hard

CH 6 · TOPIC 5 6 marks

1 (a) Sami asked 50 people which drinks they liked from tea, coffee and milk.

All 50 people like at least one of the drinks

19 people like all three drinks.

16 people like tea and coffee but do **not** like milk.

21 people like coffee and milk.

24 people like tea and milk.

40 people like coffee.

1 person likes only milk.

Sami selects at random one of the 50 people.

Work out the probability that this person likes tea.

(4 marks)

(b) Given that the person selected at random from the 50 people likes tea, find the probability that this person also likes exactly one other drink.

(2 marks)

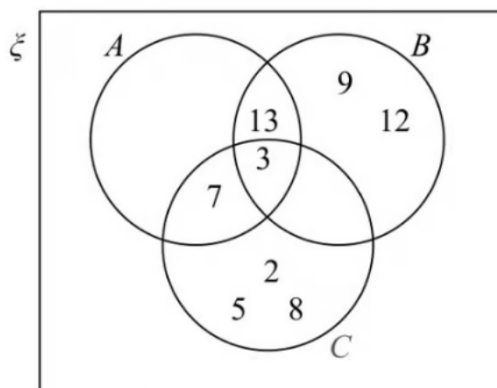
Q12**PRACTICE**

Practice • Hard

CH 6 • TOPIC 5 7 marks

11 (a) $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$ $A = \{3, 7, 11, 13\}$ $B = \{3, 6, 9, 12, 13\}$ $C = \{2, 3, 5, 6, 7, 8\}$

Complete the Venn diagram



(2 marks)

(b) List the members of the set $B' \cap C$.

(1 mark)

(c) List the members of the set $(A \cup C)' \cap B$.

(1 mark)

(d) Find $n(A' \cap B')$

(1 mark)

(e) One of the numbers in the Venn diagram is chosen at random.Find the probability that this number is in set $(A \cap B)$

(2 marks)

Q13**PRACTICE**

Practice • Hard

CH 6 • TOPIC 5

4 marks

12 (a) Each student in a group of 32 students was asked the following question.

"Do you have a desktop computer (D), a laptop (L) or a tablet (T)?"

Their answers showed that,

19 students have a desktop computer,

17 students have a laptop,

16 students have a tablet,

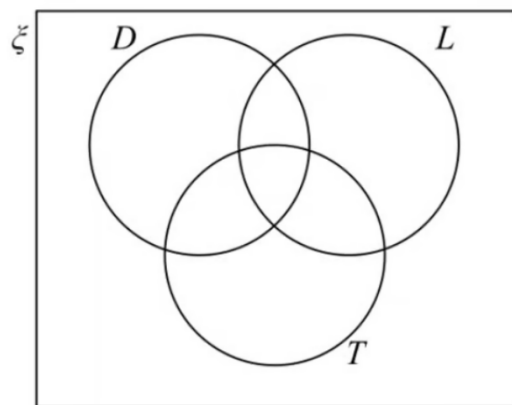
9 students have both a desktop computer and a laptop,

11 students have both a desktop computer and a tablet,

7 students have both a laptop and a tablet,

5 students have all three.

Using this information, complete the Venn diagram to show the number of students in each appropriate subset.

**(3 marks)**

(b) One of the students with both a desktop computer and a laptop is chosen at random.

Find the probability that this student also has a tablet.

(1 mark)

PAST PAPERS

Q14 PAST PAPER Edexcel 4MA1 • November 2021 • Paper 1H • Q16

CH 6 • TOPIC 5 5 marks

16 There are 32 students in a class.

In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).

25 students passed the test in Maths.

20 students passed the test in English.

14 students passed the test in French.

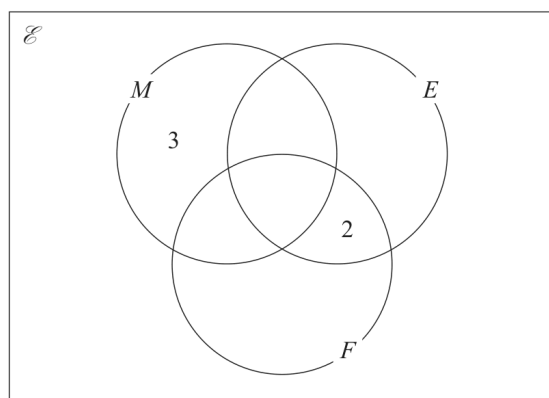
18 students passed the tests in Maths and English.

11 students passed the tests in Maths and French.

4 students failed all three tests.

 x students passed all three tests.

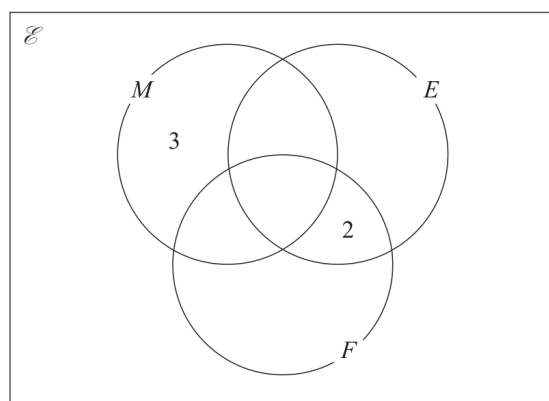
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

 $x = \dots\dots\dots$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

 $\dots\dots\dots$

(Total for Question 16 is 5 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q16

CH 6 • TOPIC 5 8 marks

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

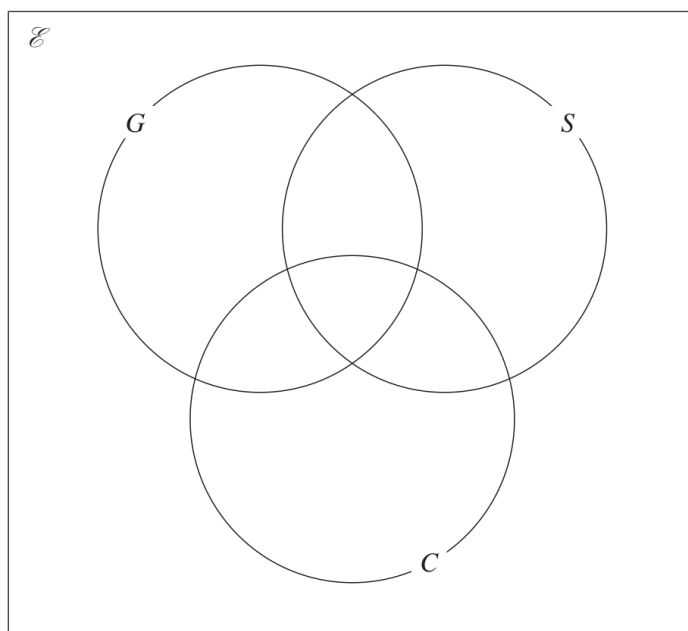
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

- (a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



- (b) Find

(i) $n(G)$

.....

(ii) $n([G \cup S]')$

.....

(iii) $n(G' \cap C)$

.....

One of the farmers who has chickens is chosen at random.

- (c) Find the probability that this farmer also has goats.

.....

(Total for Question 16 is 8 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q16

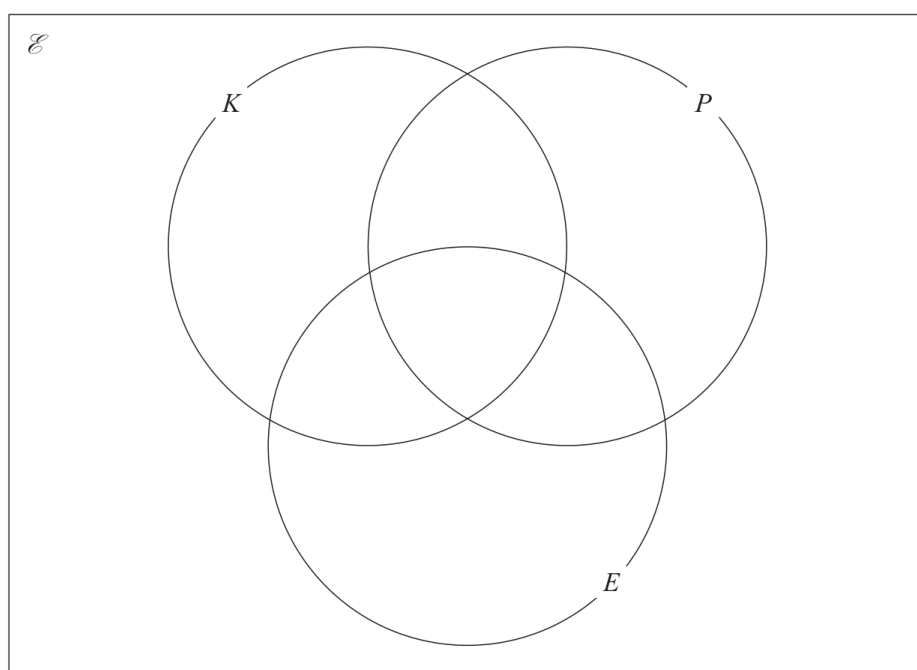
CH 6 • TOPIC 5 7 marks

- 16 60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

- 9 chose knitting, photography and embroidery
- 17 chose knitting and photography
- 16 chose photography and embroidery
- 20 chose knitting and embroidery
- 28 chose photography
- 39 chose embroidery
- 2 chose none of the workshops

- (a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



One of the students is chosen at random.

Given that this student chose photography,

- (b) find the probability that this student also chose knitting.

- (c) Find $n(P \cap K')$

- (d) Find $n([P \cup E] \cap K)$

.....

.....

.....

(Total for Question 16 is 7 marks)

Q17 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 2H • Q20

CH 6 • TOPIC 5 6 marks

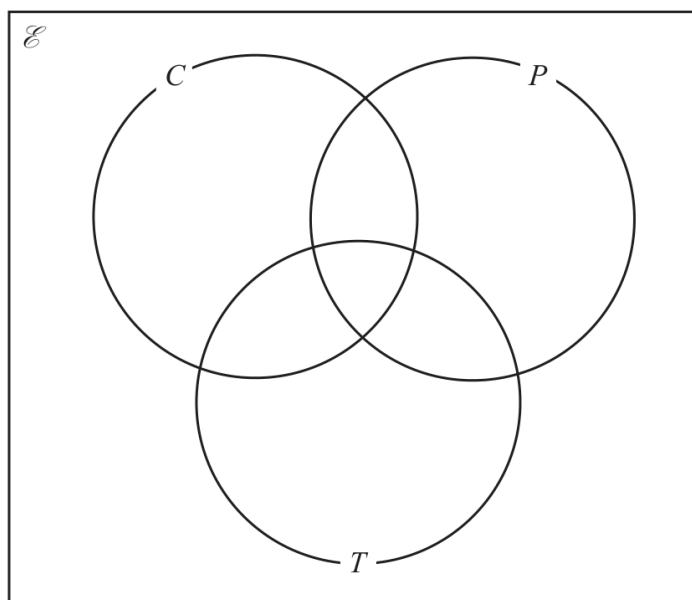
20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

- 43 grow carrots
- 12 grow carrots and potatoes and tomatoes
- 18 grow carrots and potatoes
- 27 grow carrots and tomatoes
- 32 grow potatoes and tomatoes
- 29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(b) Find $n(T' \cap C)$

.....

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

.....

(Total for Question 20 is 6 marks)

03 Tree Diagrams: Independent Events

Each pair of branches from one point adds to 1. To find the probability of a path, **multiply along** it. If several paths give what you want, **add** their answers.

WHEN YOU SEE → YOU DO

“complete the tree diagram”

→ each pair of branches adds to 1

“both ...”

→ one path: multiply

“exactly one ...”

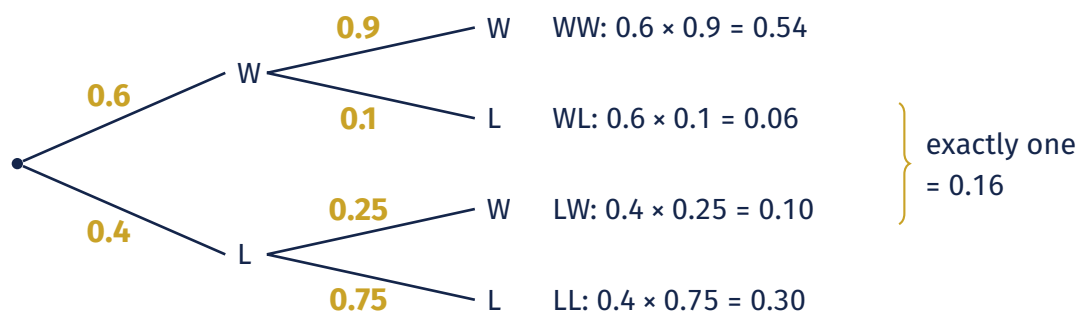
→ two paths: multiply each, then add

“at least one ...”

→ $1 - P(\text{none})$

“the same colour”

→ add the paths that match



WORKED MODEL • JANUARY 2021 P1H

Ding beats Marco with 0.6; then beats Judd with 0.9 (or 0.25 if he lost). Exactly one win?

1 WL: $0.6 \times 0.1 = 0.06$ LW: $0.4 \times 0.25 = 0.10$

two paths

ANSWER $0.06 + 0.10 = 0.16$

× **TRAP** Adding along a path. Along = $\times$; between paths = $+$.

IDEA 03

Practise: Tree Diagrams: Independent Events

My questions first, then selected practice, then real past papers.

Q18 DR ESLAM Multiply along

CH 6 • TOPIC 5 2 marks

$P(\text{rain}) = 0.3$ each day, independently. Find $P(\text{rain on both days})$ and $P(\text{rain on exactly one day})$.

SELECTED PRACTICE

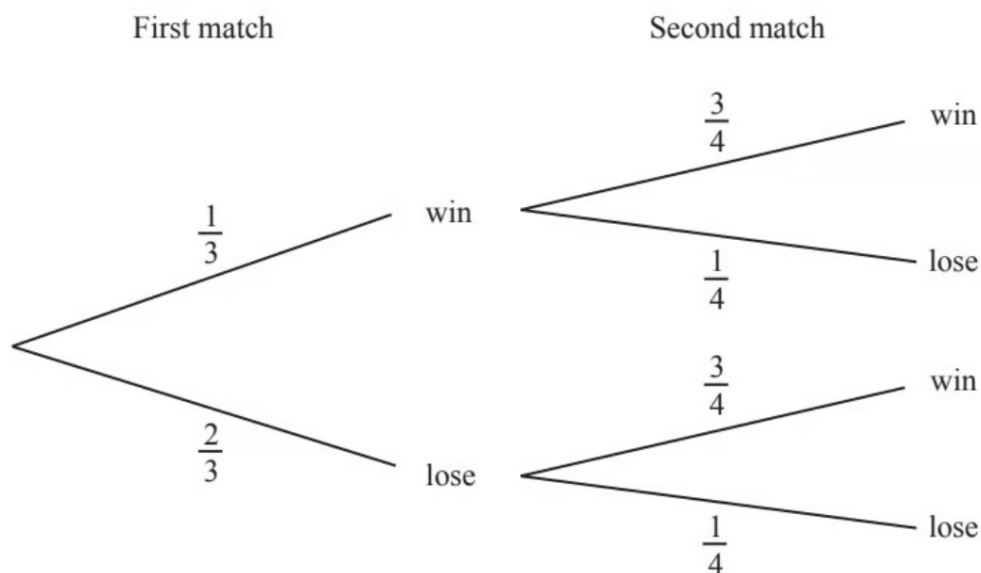
Q19**PRACTICE**

Practice • Easy

CH 6 • TOPIC 5 3 marks

10 A soccer team plays two matches.

The tree diagram shows the probability of the team winning or losing the matches.



Find the probability that the soccer team wins at least one of the two matches.

(3 marks)

Q20**PRACTICE**

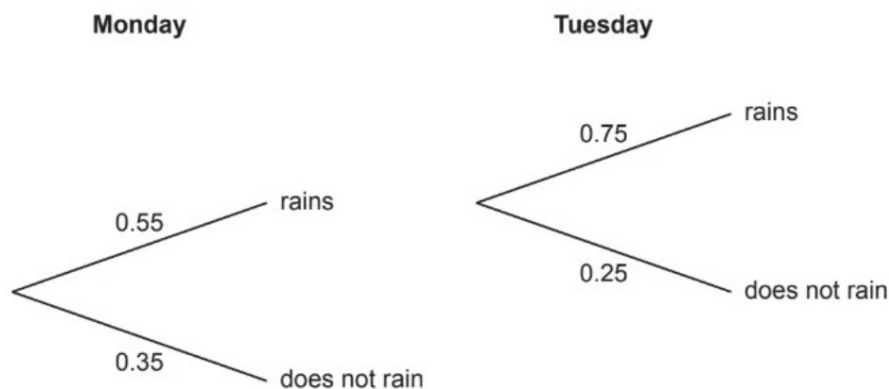
Practice • Medium

CH 6 • TOPIC 5 3 marks

12 A weather forecast says

- the probability that it will rain on Monday is 0.55 and
- the probability that it will rain on Tuesday is 0.25.

Ella draws a tree diagram to show this information.



Write down three errors that Ella has made with her tree diagram.

(3 marks)

Q21**PRACTICE**

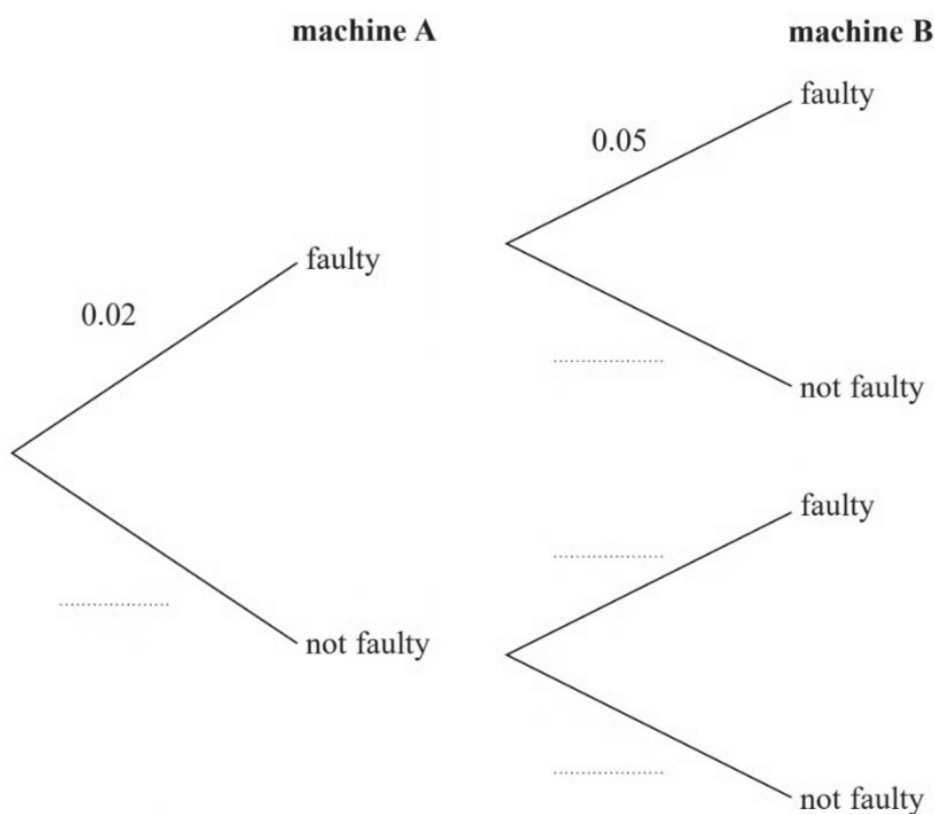
Practice • Hard

CH 6 • TOPIC 5 5 marks

5 (a) Machine A and machine B make bottles.

The probability that a bottle made by machine A is faulty is 0.02 The probability that a bottle made by machine B is faulty is 0.05

Complete the probability tree diagram.

**(2 marks)**

(b) Shazia takes at random one bottle made by machine A and one bottle made by machine B.

Work out the probability that at least one of these bottles is faulty.

(3 marks)

Q22**PRACTICE**

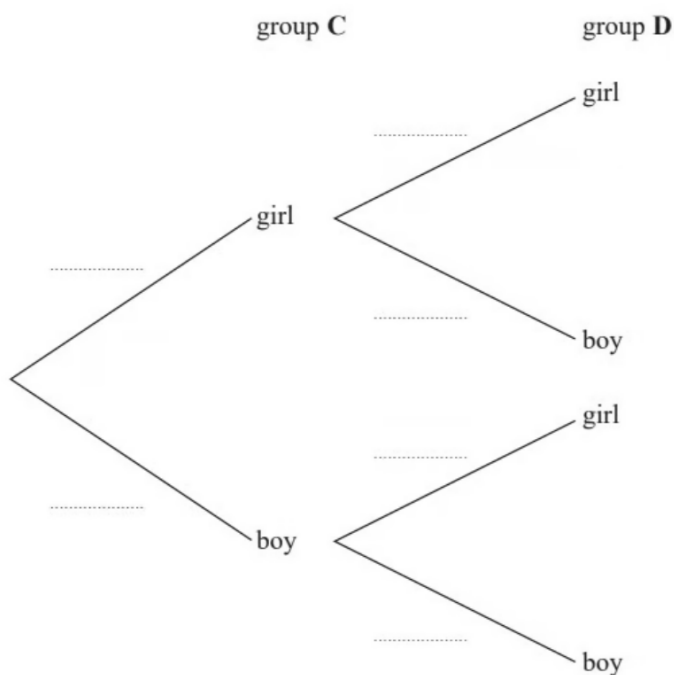
Practice • Very Hard

CH 6 • TOPIC 5 7 marks

- 9 (a) In group **C**, there are 6 girls and 8 boys.
In group **D**, there are 3 girls and 7 boys.

A team is made by picking at random one child from group **C** and one child from group **D**.

Complete the probability tree diagram.

**(2 marks)**

- (b) Work out the probability that there are two boys in the team.

(2 marks)

- (c) After the first team has been picked, a second team is picked. One child is picked at random from the children left in group **C** and one child is picked at random from the children left in group **D**.

Work out the probability that there are two boys in each of the two teams.

(3 marks)

PAST PAPERS

Q23 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 2HR • Q16

CH 6 • TOPIC 5 8 marks

16 Cody has two bags of counters, bag **A** and bag **B**.

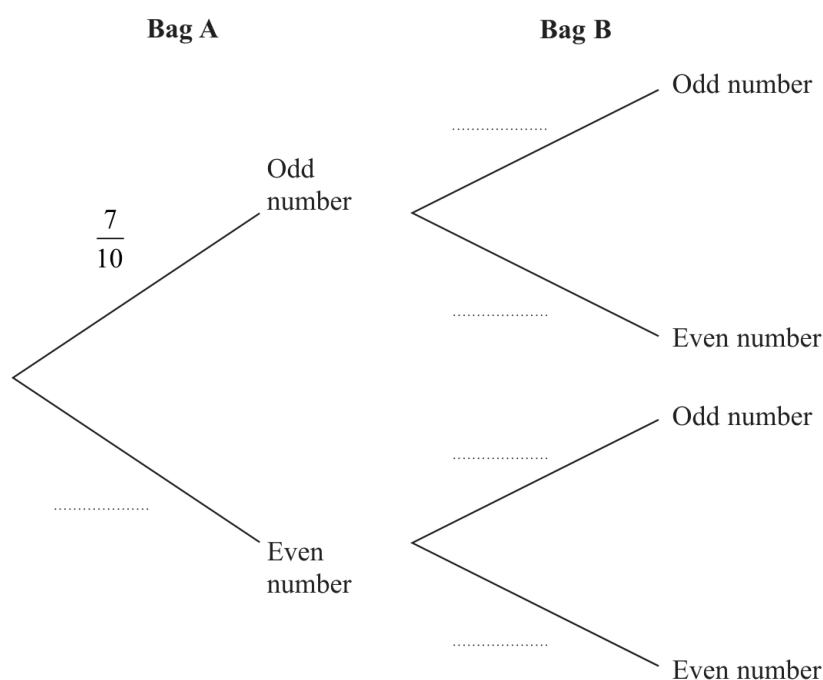
Each of the counters has either an odd number or an even number written on it.

There are 10 counters in bag **A** and 7 of these counters have an **odd** number written on them.

There are 12 counters in bag **B** and 7 of these counters have an **odd** number written on them.

Cody is going to take at random a counter from bag **A** and a counter from bag **B**.

(a) Complete the probability tree diagram.



(b) Calculate the probability that the total of the numbers on the two counters will be an odd number.

.....

Harriet also has a bag of counters.

Each of her counters also has either an odd number or an even number written on it.

Harriet is going to take at random a counter from her bag of counters.

The probability that the number on each of Cody's two counters **and** the number on

Harriet's counter will all be even is $\frac{3}{100}$

(c) Find the least number of counters that Harriet has in her bag.
Show your working clearly.

.....

(Total for Question 16 is 8 marks)

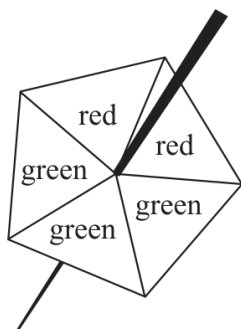
Q24

PAST PAPER

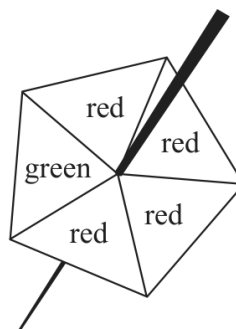
Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q15

CH 6 • TOPIC 5 5 marks

15 Harry has two fair 5-sided spinners.



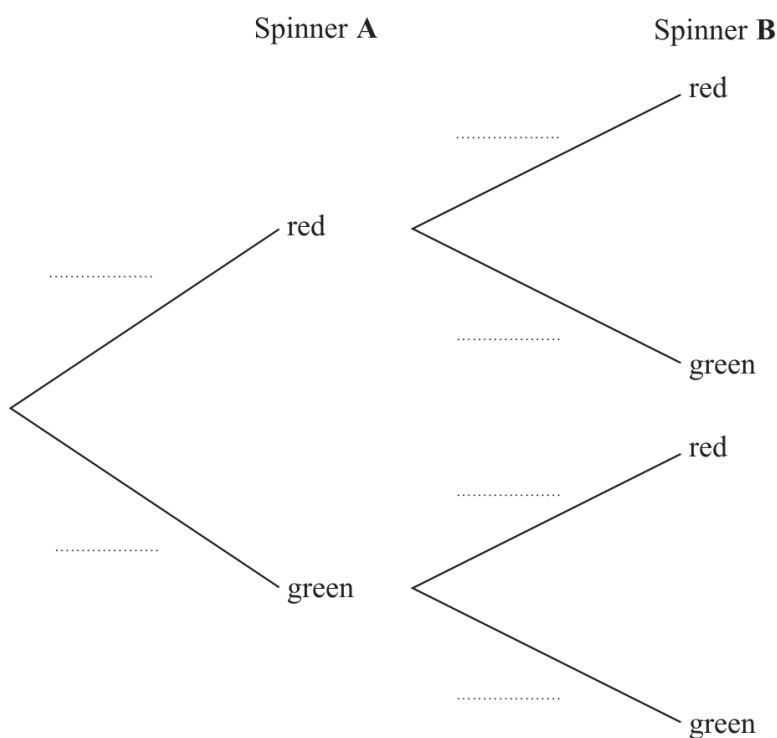
Spinner A



Spinner B

Harry is going to spin each spinner once.

(a) Complete the probability tree diagram.



(b) Work out the probability that at least one of the spinners will land on green.

(Total for Question 15 is 5 marks)

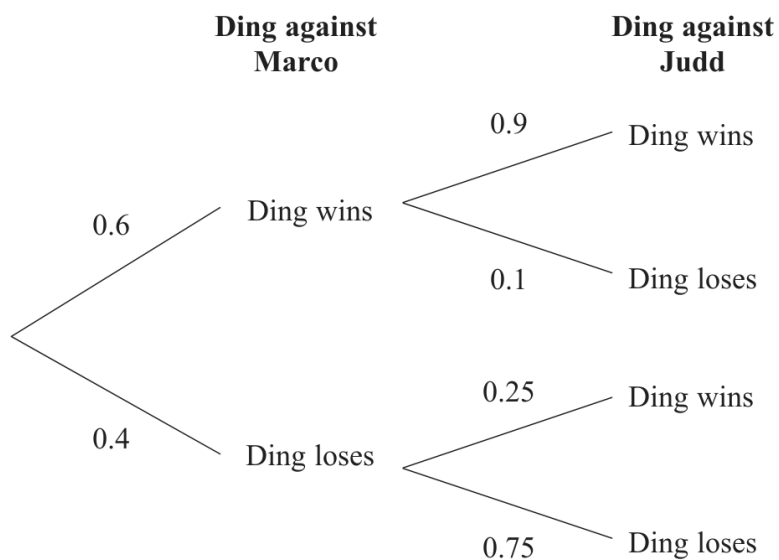
Q25**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q14

CH 6 • TOPIC 5 5 marks

- 14 Ding is going to play one game of snooker against each of two of his friends, Marco and Judd.

The probability tree diagram gives information about the probabilities that Ding will win or lose each of these two games.



- (a) Work out the probability that Ding will win both games.

.....

- (b) Work out the probability that Ding will win exactly one of the games.

.....

(Total for Question 14 is 5 marks)

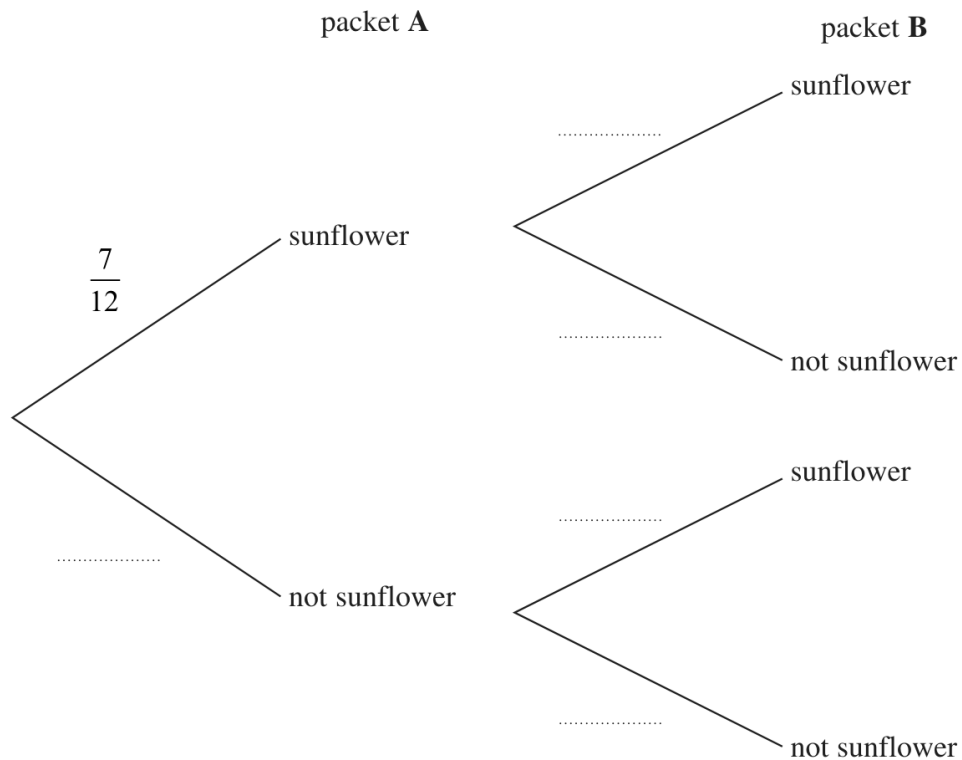
14 Aika has 2 packets of seeds, packet **A** and packet **B**

There are 12 seeds in packet A and 7 of these are sunflower seeds.

There are 15 seeds in packet **B** and 8 of these are sunflower seeds.

Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.



(b) Calculate the probability that Aika will take two sunflower seeds.

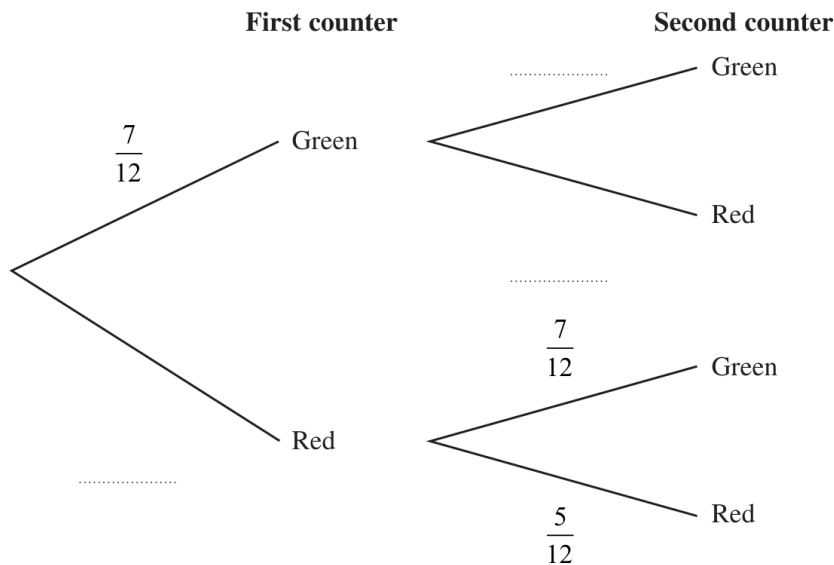
(Total for Question 14 is 4 marks)

- 13** Hector has a bag that contains 12 counters.
There are 7 green counters and 5 red counters in the bag.

Hector takes at random a counter from the bag.
He looks at the counter and puts the counter back into the bag.

Hector then takes at random a second counter from the bag.
He looks at the counter and puts the counter back into the bag.

- (a) Complete the probability tree diagram.



- (b) Work out the probability that both counters are red.

Meghan has a jar containing 15 counters.
There are only blue counters, green counters and red counters in the jar.

Hector is going to take at random one of the counters from his bag of 12 counters. He will look at the counter and put the counter back into the bag.

Hector is then going to take at random a second counter from his bag. He will look at the counter and put the counter back into the bag.

Meghan is then going to take at random one of the counters from her jar of counters. She will look at the counter and put the counter back into the jar.

The probability that the 3 counters each have a different colour is $\frac{7}{24}$

- (c) Work out how many blue counters there are in the jar.

(Total for Question 13 is 7 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q13

CH 6 • TOPIC 5 4 marks

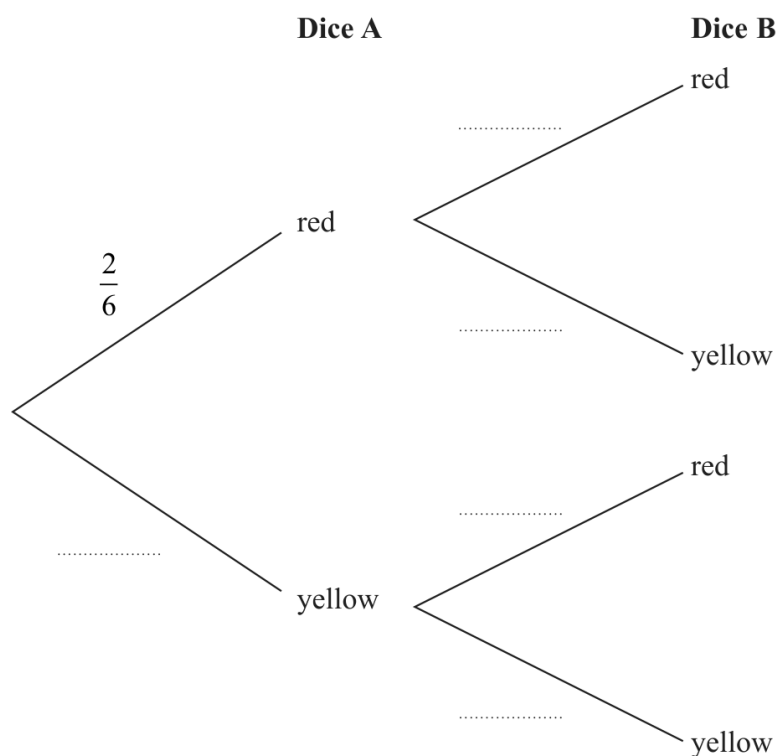
13 Narin has two fair 6-sided dice.

Dice **A** has 2 red faces and 4 yellow faces.

Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.



(b) Work out the probability that both dice land on yellow.

.....

(Total for Question 13 is 4 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q11

CH 6 • TOPIC 5 5 marks

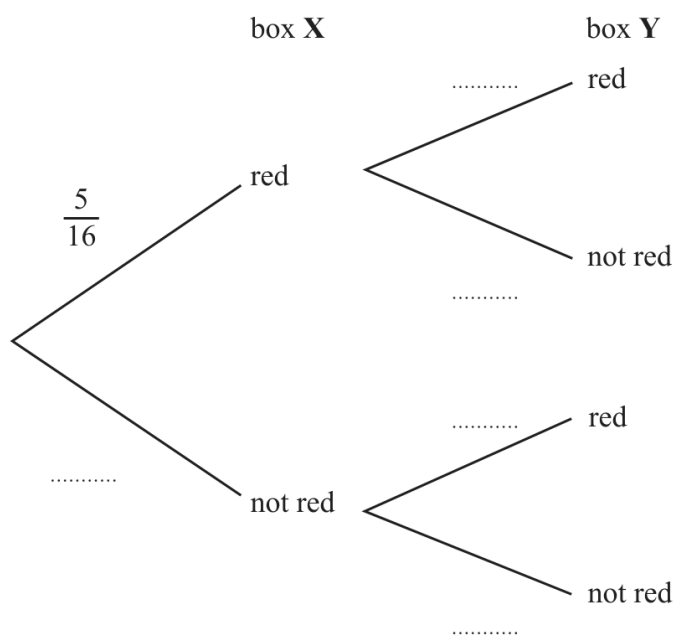
11 Sid has 2 boxes of crayons, box **X** and box **Y**

5 of the 16 crayons in box **X** are red.

7 of the 20 crayons in box **Y** are red.

Sid takes at random one crayon from box **X** and one crayon from box **Y**

(a) Complete the probability tree diagram.



(b) Work out the probability that Sid takes two crayons that are red or two crayons that are not red.

(Total for Question 11 is 5 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q13

CH 6 • TOPIC 5 5 marks

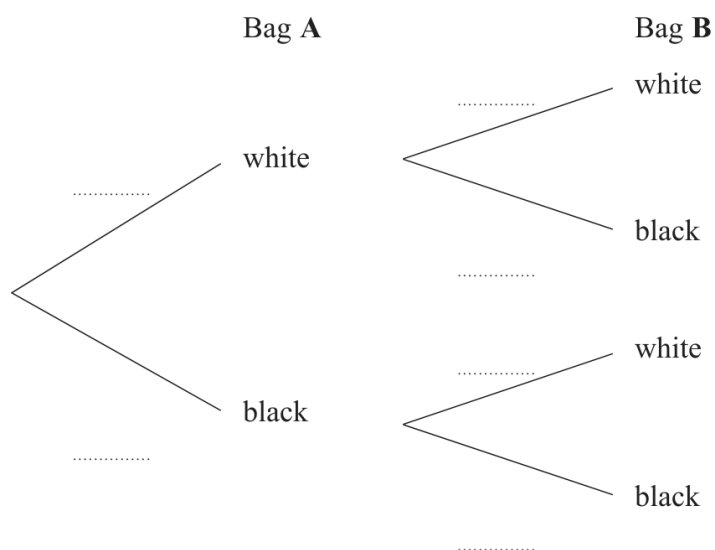
13 Harman has two bags of beads.

In bag **A**, there are 3 white beads and 7 black beads.

In bag **B**, there are 5 white beads and 4 black beads.

Harman takes at random a bead from bag **A** and a bead from bag **B**

(a) Complete the probability tree diagram.



(b) Work out the probability that Harman takes two beads of the same colour.

(Total for Question 13 is 5 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q14

CH 6 • TOPIC 5 4 marks

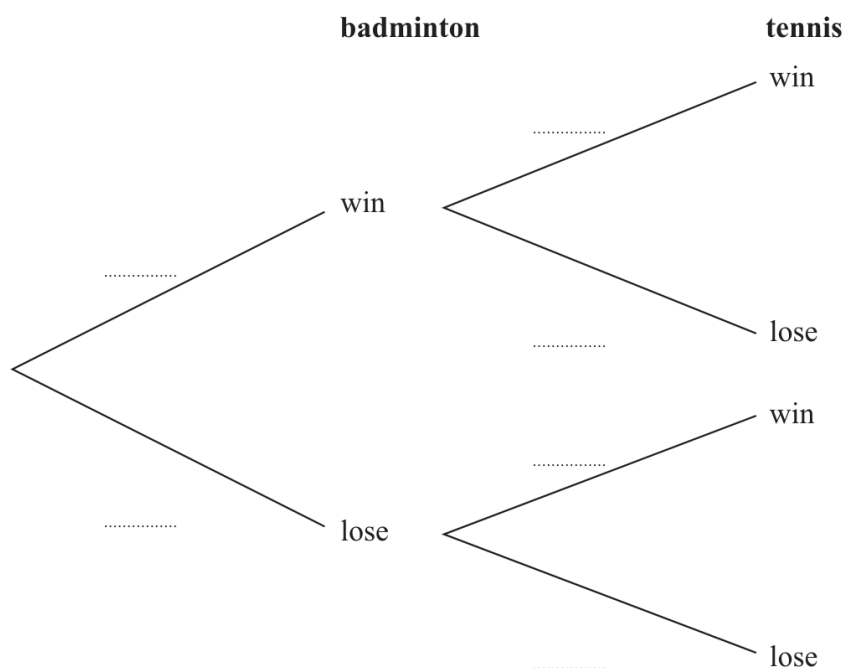
14 Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7

The probability that he will win the game of tennis is 0.4

(a) Complete the probability tree diagram.



(b) Work out the probability that Ricardo loses both games.

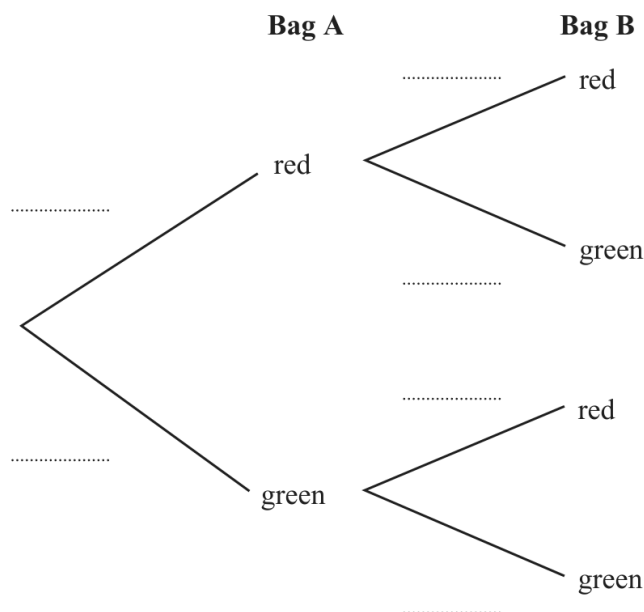
(Total for Question 14 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q14

CH 6 • TOPIC 5 7 marks**14** Sara has two bags, **A** and **B**In bag **A**, there are only 5 red beads and 4 green beads.In bag **B**, there are only 7 red beads and 3 green beads.Sara takes at random a bead from bag **A**She then takes at random a bead from bag **B**

(a) Use this information to complete the probability tree diagram.



(b) Work out the probability that Sara takes two red beads.

.....

Sara puts the beads back into their original bags.

Sara also has a box of beads.

In the box, there are only red beads and green beads.

When a bead is taken at random from the box, the probability that it is

a green bead is $\frac{2}{11}$ Sara takes at random a bead from bag **A**She then takes at random a bead from bag **B**

She then takes at random a bead from the box.

(c) Work out the probability that Sara takes more red beads than green beads.

.....

(Total for Question 14 is 7 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q13

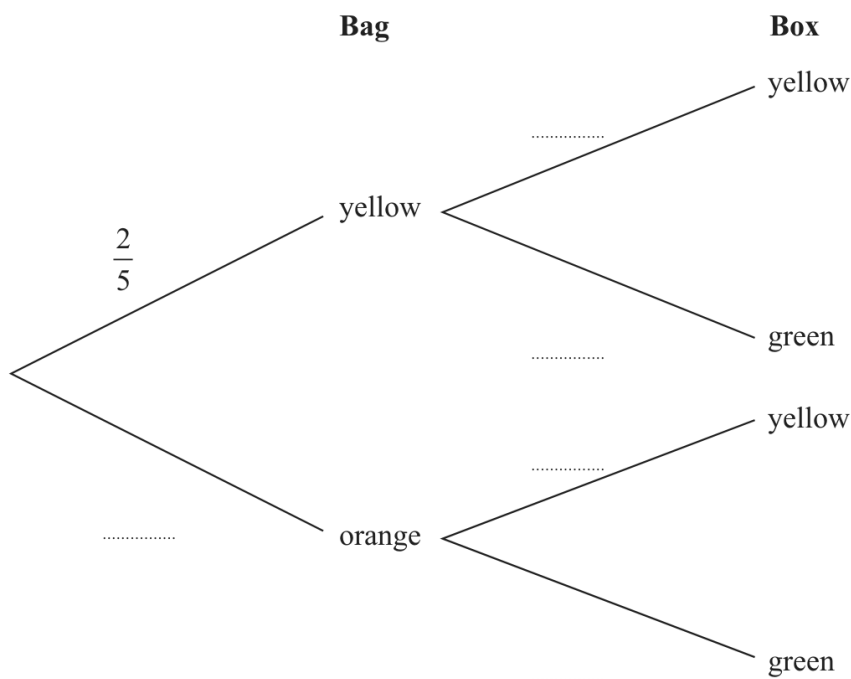
CH 6 • TOPIC 5 5 marks

13 In a bag, there are only 2 yellow beads and 3 orange beads.

In a box, there are only 3 yellow beads and 5 green beads.

Chi takes at random a bead from the bag and a bead from the box.

(a) Complete the probability tree diagram.



(b) Work out the probability that Chi takes two beads of different colours.

(Total for Question 13 is 5 marks)

04 Tree Diagrams: Without Replacement and Conditional Branches

If the first item is **not put back**, the second set of branches changes: one fewer on the bottom, and one fewer on top for the colour already taken. If the question says “if ... then ...”, each branch pair gets its own numbers.

WHEN YOU SEE → YOU DO

“takes two glasses”, “does not replace”

→ second fraction: top -1 (if same kind), bottom -1

“if he wins the first game, the probability ... is 0.9”

→ that number goes on the branch after “win”

“given that it rains, the match is cancelled with 0.7”

→ 0.7 after “rain”, 0.3 next to it

“at least one yellow”

→ $1 - P(\text{no yellow})$

WORKED MODEL • JANUARY 2020 P2H

13 large and 7 small glasses. Two are taken at random. $P(\text{two small})$?

1 first small: $\frac{7}{20}$; second small: $\frac{6}{19}$

one small gone

ANSWER $\frac{7}{20} \times \frac{6}{19} = \frac{42}{380} = \frac{21}{190}$

× **TRAP** Keeping $\frac{7}{20}$ for the second pick. Without replacement, the bottom drops to 19.

IDEA 04

Practise: Tree Diagrams: Without Replacement and Conditional Branches

My questions first, then selected practice, then real past papers.

Q34 **DR ESLAM** No replacement

CH 6 • TOPIC 5 3 marks

A bag has 4 red and 6 blue beads. Two are taken without replacement. Find $P(\text{both blue})$.

SELECTED PRACTICE

Q35

PRACTICE

Practice • Medium

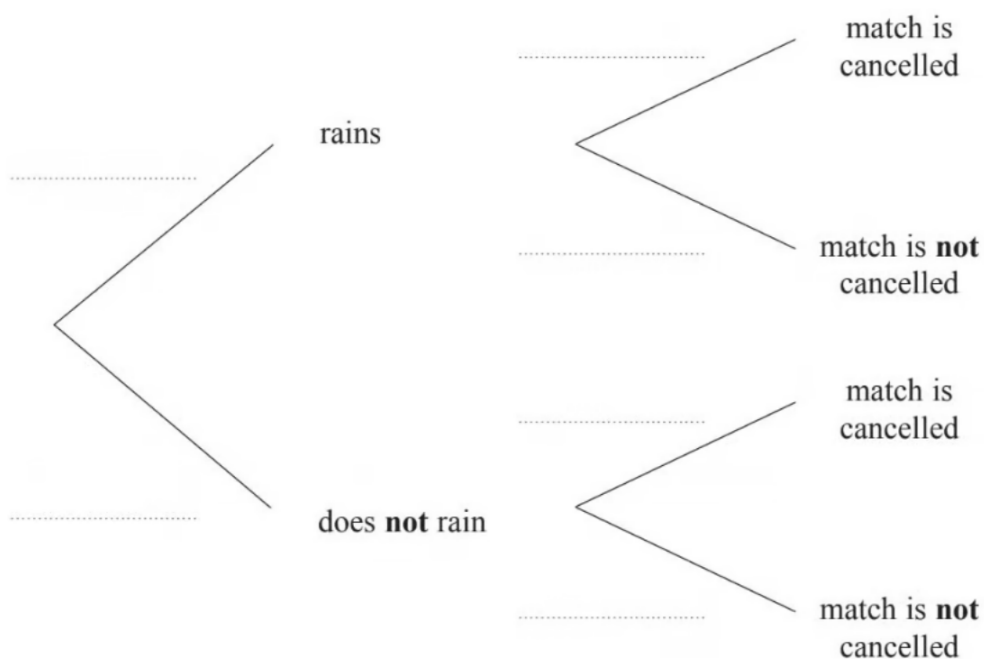
CH 6 • TOPIC 5 5 marks

- 6 (a) The probability that it will rain on a day in June is 0.2

When it rains the probability that my tennis match is cancelled is 0.7

When it does **not** rain, the probability that my tennis match is **not** cancelled is 0.95

Complete the probability tree diagram for this information.



(3 marks)

- (b) Work out the probability that, on a day in June, it does **not** rain and my tennis match is cancelled.

(2 marks)

Q36**PRACTICE**

Practice • Hard

CH 6 • TOPIC 5 5 marks

8 (a) Yvonne has 10 tulip bulbs in a bag.

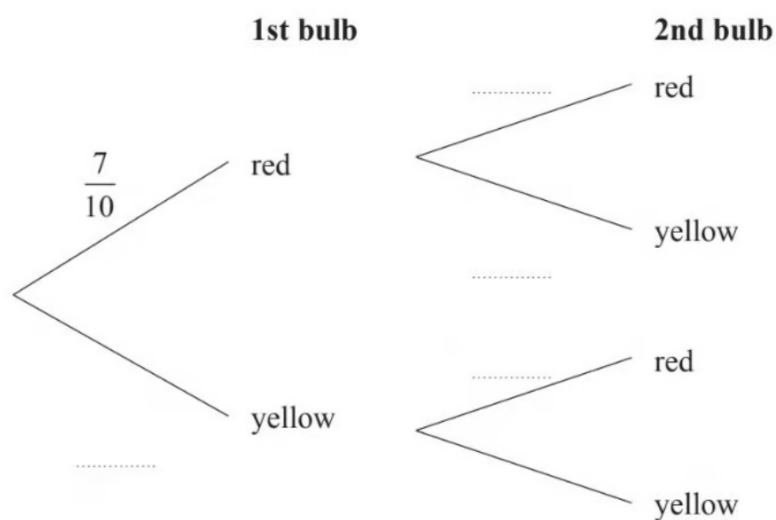
7 of the tulip bulbs will grow into red tulips.

3 of the tulip bulbs will grow into yellow tulips.

Yvonne takes at random two tulip bulbs from the bag.

She plants the bulbs.

Complete the probability tree diagram.

**(2 marks)**

(b) Work out the probability that at least one of the bulbs will grow into a yellow tulip.

(3 marks)

Q37**PRACTICE**

Practice · Very Hard

CH 6 • TOPIC 5 5 marks

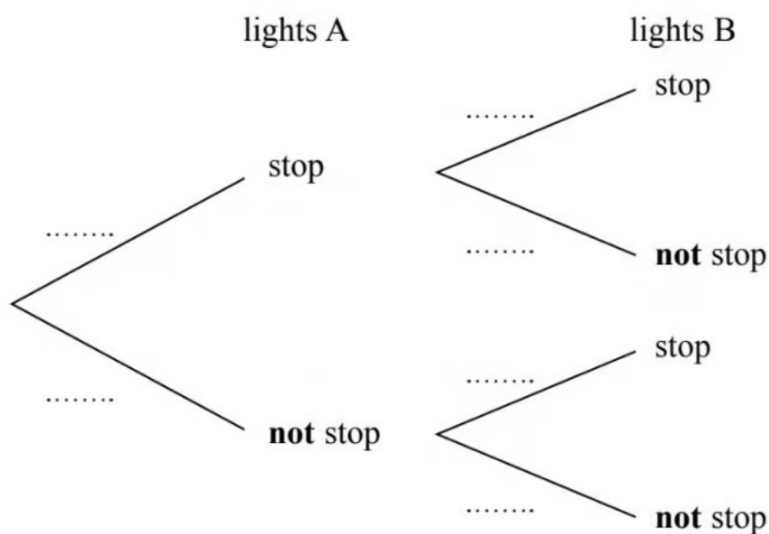
2 (a) A and B are two sets of traffic lights on a road.

The probability that a car is stopped by lights A is 0.4

If a car is stopped by lights A, then the probability that the car is **not** stopped by lights B is 0.7

If a car is **not** stopped by lights A, then the probability that the car is **not** stopped by lights B is 0.2

Complete the probability tree diagram for this information.

**(2 marks)**

(b) Mark drove along this road.

He was stopped by just one of the sets of traffic lights.

Is it more likely that he was stopped by lights A or by lights B?

You must show your working.

(3 marks)

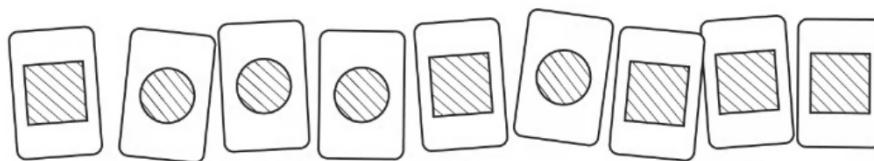
Q38

PRACTICE

Practice • Very Hard

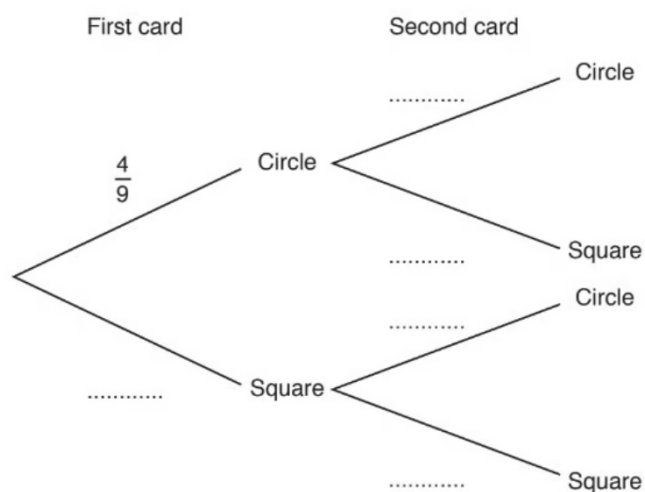
CH 6 • TOPIC 5 5 marks

12 (a) Reuben is playing a matching game with these cards.



He turns the cards over and shuffles them. Reuben takes a card and keeps it. He then takes a second card. If the cards are different, he wins the game.

Complete this tree diagram to show the probabilities for each card picked in the game.



(2 marks)

(b) What is the probability that Reuben wins the game?

(3 marks)

PAST PAPERS

Q39**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q14

CH 6 • TOPIC 5 4 marks

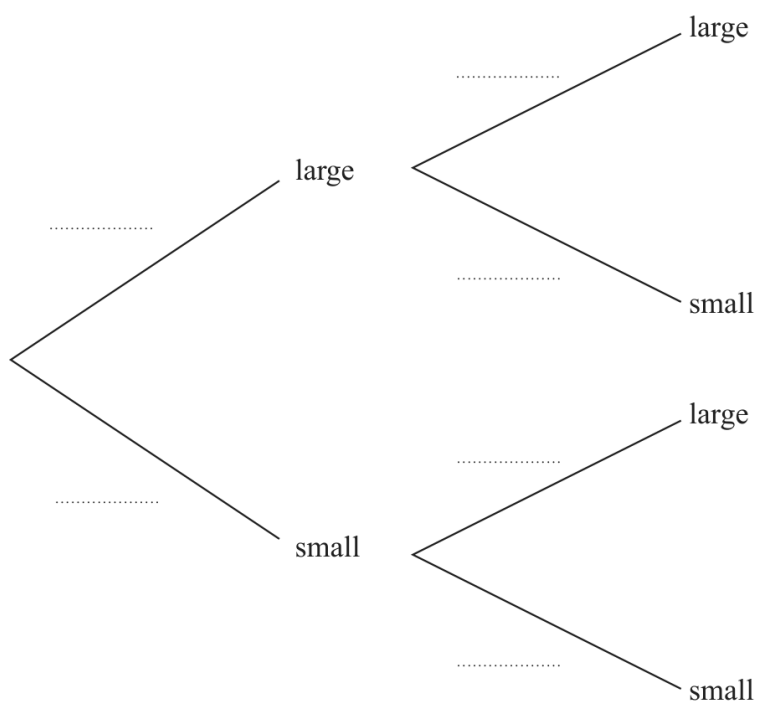
14 There are 20 glasses in a cupboard.

13 of the glasses are large

7 of the glasses are small

Roberto takes at random two glasses from the cupboard.

(a) Complete the probability tree diagram.



(b) Work out the probability that Roberto takes two small glasses.

(Total for Question 14 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q12

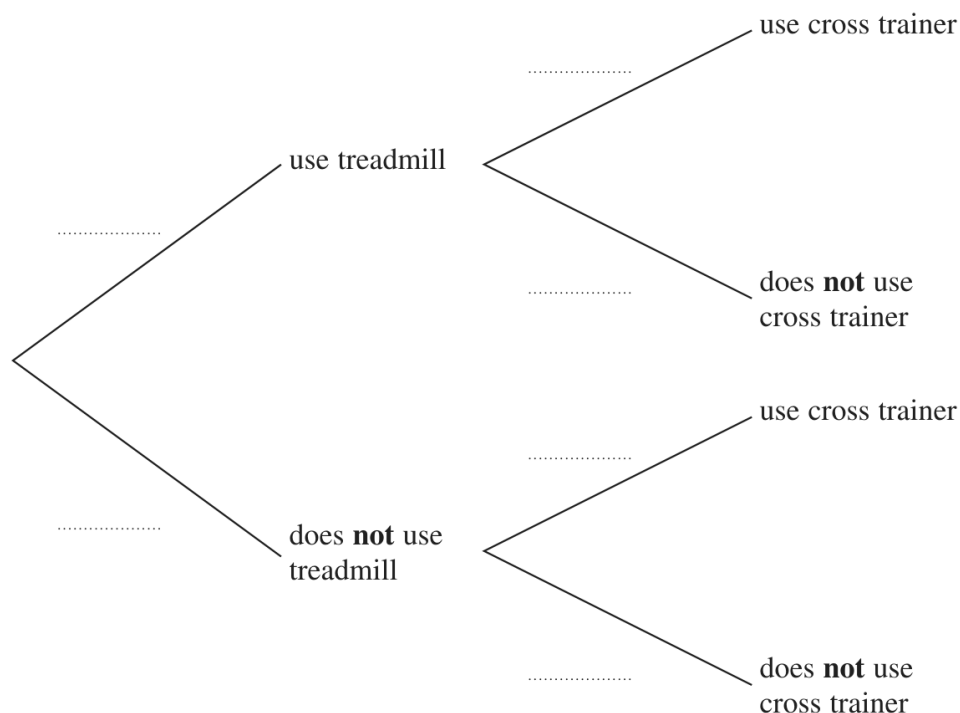
CH 6 • TOPIC 5 4 marks**12** Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3

When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.



(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

(Total for Question 12 is 4 marks)

Q41 PAST PAPER Edexcel 4MA1 • May 2023 • Paper 1HR • Q13

CH 6 • TOPIC 5 3 marks

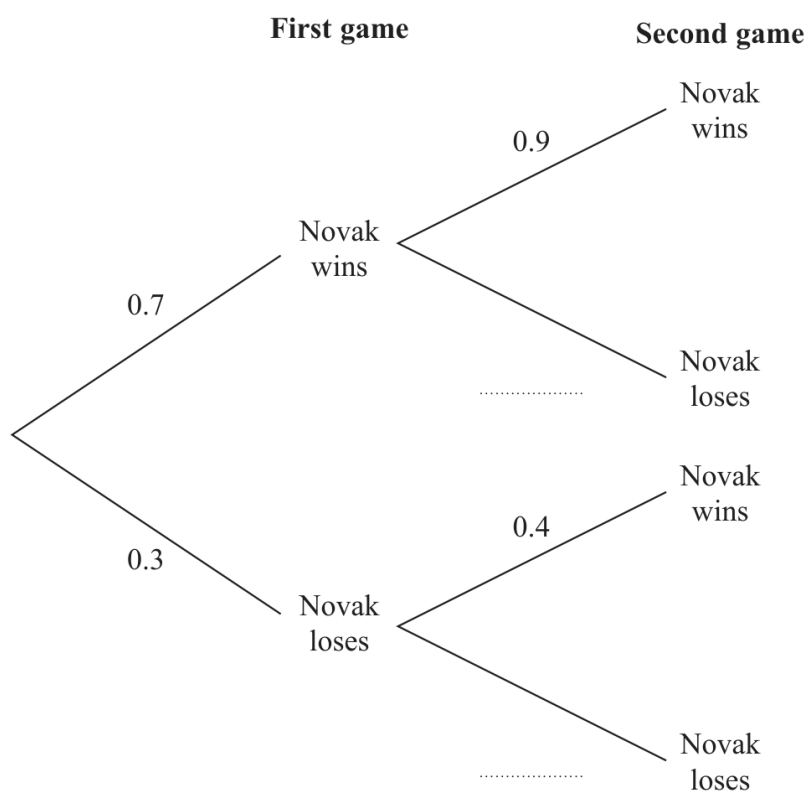
13 Novak is going to play two games of tennis.

The probability that he wins the first game is 0.7

If he wins the first game, the probability that he wins the second game is 0.9

If he loses the first game, the probability that he wins the second game is 0.4

(a) Complete the probability tree diagram.



(b) Work out the probability that Novak wins both games of tennis.

(Total for Question 13 is 3 marks)

Q42

PAST PAPER

Edexcel 4MA1 • November 2024 • Paper 1H • Q13

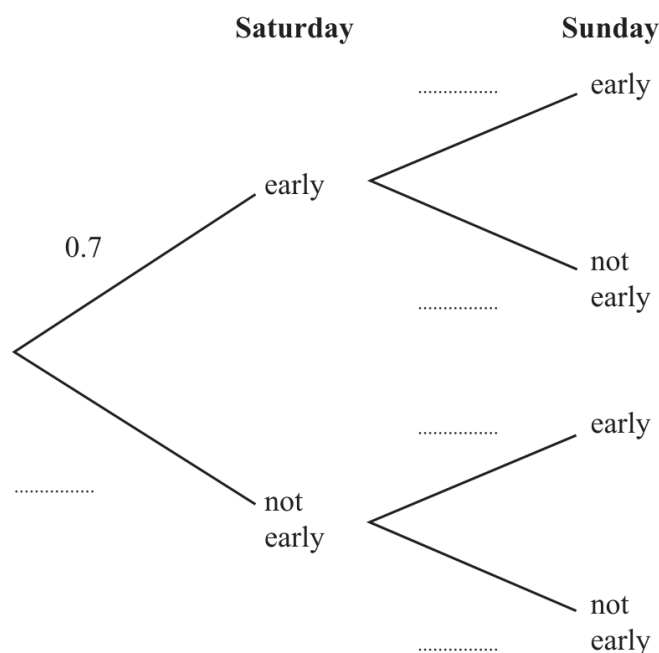
CH 6 • TOPIC 5 4 marks

13 The probability that Thomas gets to work early on Saturday is 0.7

If Thomas gets to work early on Saturday, the probability that he will get to work early on Sunday is 0.9

If Thomas does **not** get to work early on Saturday, the probability that he will get to work early on Sunday is 0.6

(a) Use this information to complete the probability tree diagram.



(b) Work out the probability that Thomas gets to work early on both Saturday and Sunday.

.....
(Total for Question 13 is 4 marks)

05 Longer Trees: Three Stages and More Branches

Three games, or three outcomes per game? The rules do not change: multiply along a path, add the paths you want. List the paths first so you do not miss one.

WHEN YOU SEE → YOU DO

win / draw / lose

→ three branches from each point, still adding to 1

“the same number of points after 2 games”

→ list: DD, WL, LW

a third race depends on the first two

→ extend only the paths the question needs

“at most 2 attempts”

→ pass first time, or fail then pass

WORKED MODEL • NOVEMBER 2021 P1H

Magnus wins $\frac{2}{9}$, draws $\frac{4}{9}$, loses $\frac{3}{9}$. $P(\text{same points after 2 games})?$

1 paths: DD, WL, LW

list first

$$2 \quad \frac{4}{9} \cdot \frac{4}{9} + \frac{2}{9} \cdot \frac{3}{9} + \frac{3}{9} \cdot \frac{2}{9} = \frac{16}{81} + \frac{6}{81} + \frac{6}{81}$$

multiply, then add

ANSWER $\frac{28}{81}$

IDEA 05

Practise: Longer Trees: Three Stages and More Branches

My questions first, then selected practice, then real past papers.

Q43 **DR ESLAM** At least one

CH 6 • TOPIC 5 2 marks

$P(\text{hit}) = 0.6$ for each of three shots, independently. Find $P(\text{at least one hit})$.

SELECTED PRACTICE

Q44 **PRACTICE** Practice • Hard

CH 6 • TOPIC 5 3 marks

- 27** Sophie takes an examination.
If she fails the examination, she will resit.

The probability that Sophie passes the examination on her first attempt is 0.7
If she fails the examination on any attempt, the probability she passes on the next attempt is 0.9

Work out the probability that Sophie takes at most 2 attempts to pass the examination.

(3 marks)

Q45**PRACTICE**

Practice · Hard

CH 6 · TOPIC 5 4 marks

28 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag. He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents

(4 marks)

Q46**PRACTICE**

Practice • Very Hard

CH 6 • TOPIC 5 6 marks

10 (a) Anna plays a game with an ordinary, fair dice.

If she rolls 1 she wins.

If she rolls 2 or 3 she loses.

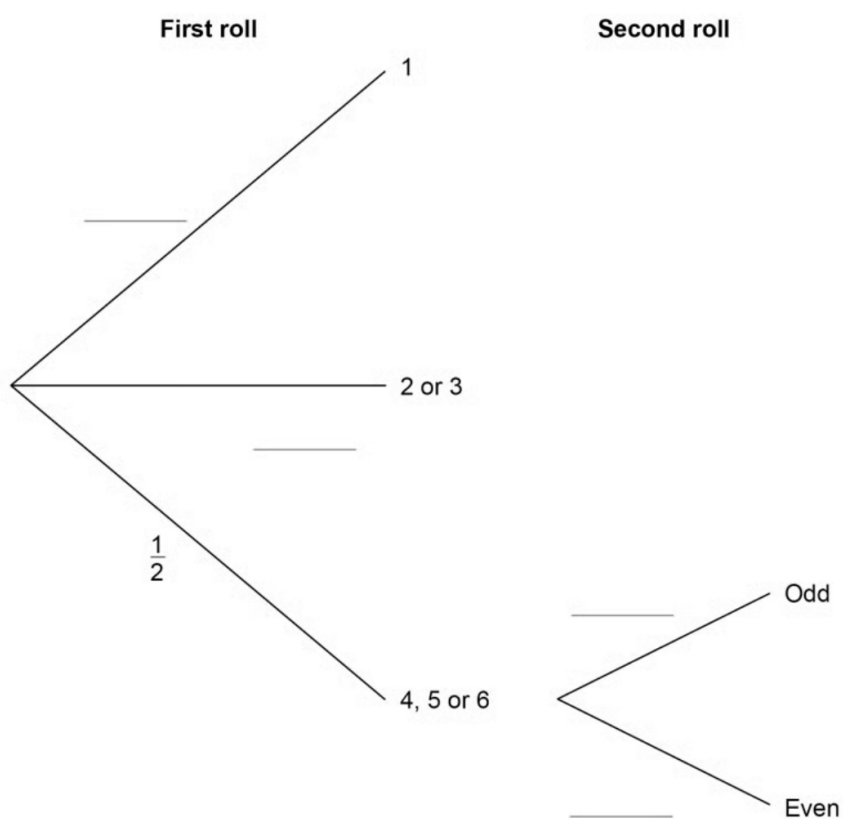
If she rolls 4, 5 or 6 she rolls again.

When she has to roll again,

if she rolls an odd number she wins

if she rolls an even number she loses.

Complete the tree diagram with the four missing probabilities.

**(2 marks)**

(b) Is Anna more likely to win or to lose?

You **must** work out the probability that she wins.

(4 marks)

Q47

PRACTICE

Practice • Very Hard

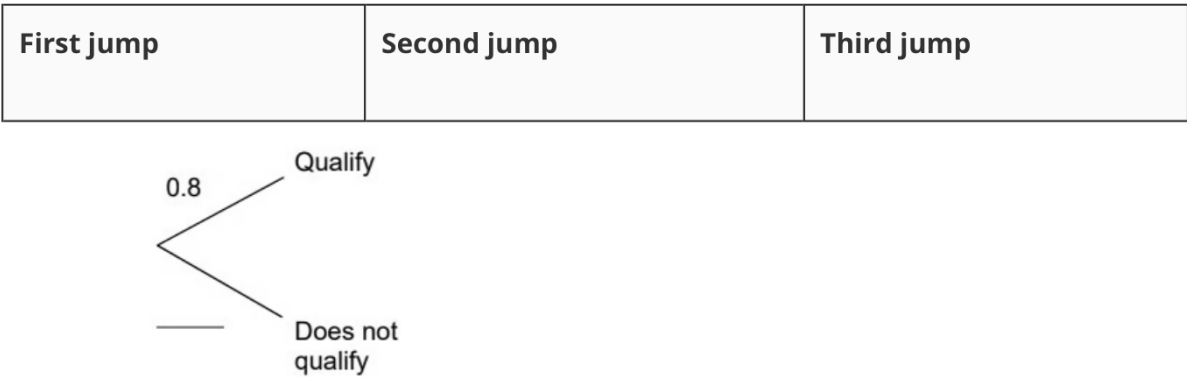
CH 6 • TOPIC 5

4 marks

PRACTISE • 6.5

- 11 (a) On Friday, Greg takes part in a long jump competition. He has to jump at least 7.5 metres to qualify for the final on Saturday.
- He has up to three jumps to qualify.
 - If he jumps at least 7.5 metres he does **not** jump again on Friday.

Each time Greg jumps, the probability he jumps at least 7.5 metres is 0.8
Assume each jump is independent. Complete the tree diagram.



(2 marks)

- (b) Work out the probability that he does **not** need the third jump to qualify.

(2 marks)

PAST PAPERS

Q48**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q13

CH 6 • TOPIC 5 8 marks

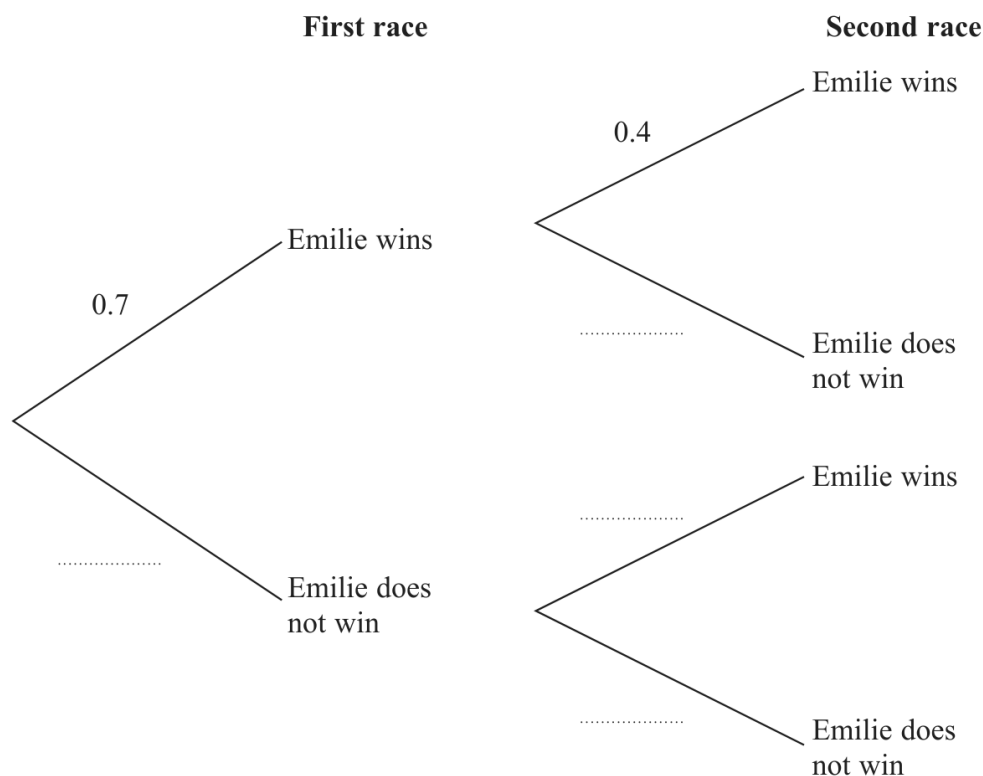
13 Emilie takes part in two races.

The probability that she wins the first race is 0.7

The probability that she wins the second race is 0.4

The outcomes of the two races are independent.

(a) Complete the probability tree diagram.



(b) Work out the probability that Emilie wins exactly one of the two races.

.....

Emilie is going to take part in a third race.

If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

.....

(Total for Question 13 is 8 marks)

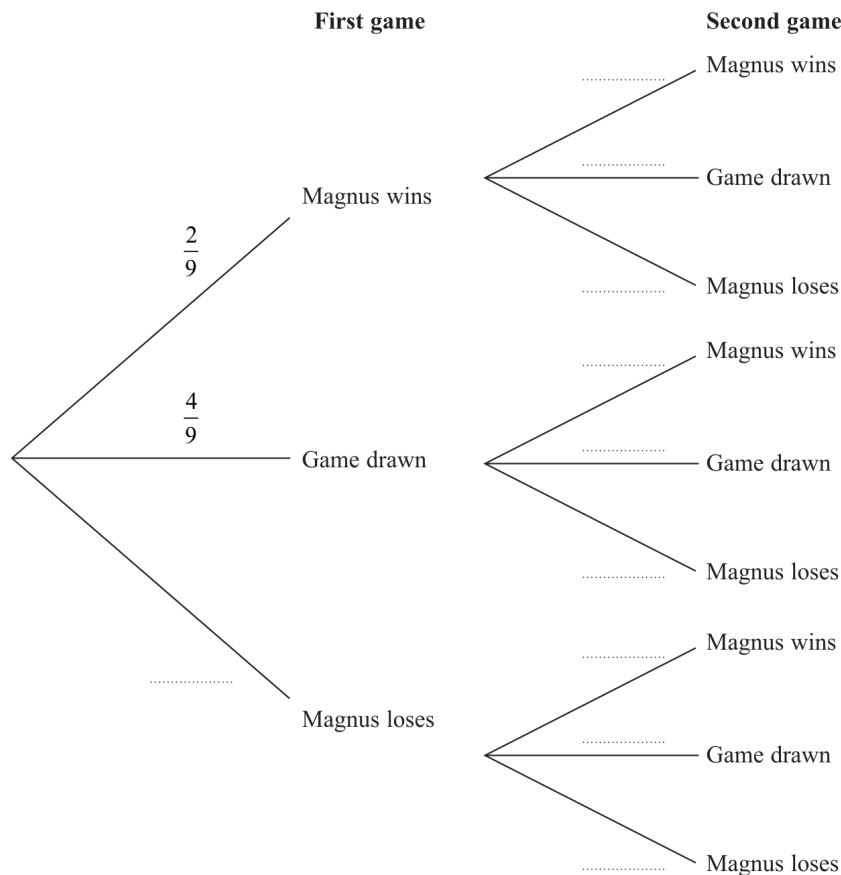
15 Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

(a) Complete the probability tree diagram.



For each game of chess,

the winner gets 2 points and the loser gets 0 points, when the game is drawn, each player gets 1 point.

(b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

(Total for Question 15 is 8 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|--|
| 01 | Two circles | Overlap first; “given” → smaller bottom. |
| 02 | Three circles | Centre, pairs – centre, singles, outside. |
| 03 | Independent trees | Branches add to 1; × along, + between. |
| 04 | Dependent trees | No replacement: one fewer; “if ... then” branches. |
| 05 | Longer trees | List the paths; at least one = 1 – none. |

BOTTOM LINE Venn: from the middle out • Tree: multiply along, add between • at least one = $1 - P(\text{none})$.

Answers

Venn and Tree Diagrams • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • TWO-CIRCLE VENN DIAGRAMS

- Q1** Dr Eslam • Fill the regions A only 16, both 6, B only 9, outside 9; $P = \frac{9}{40}$
- Q2** Practice • Easy (a) A only 1, 5, 8; both 6, 9; B only 2; outside 3, 4, 7 (b) $\frac{2}{9}$
- Q3** Practice • Medium (a)(i) 10, 12, 14, 15, 16, 18 (ii) 12, 18 (b) $\frac{7}{10}$
- Q4** Practice • Hard (a) $\frac{8}{21}$ (b) $\frac{4}{15}$ (c) $\frac{28}{36} > \frac{25}{36}$
- Q5** Practice • Hard both = $85 + 78 - 100 = 63$; football only 22:
 $\frac{22}{85}$
- Q6** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q1 $\{2, 4, 6, 12\}, \{5, 7, 8, 9, 10, 11, 13, 14\}, \frac{3}{14}$
- Q7** Edexcel 4MA1 • January 2021 • Paper 2HR • Q16 $\frac{3}{10}, \frac{37}{40}$
- Q8** Edexcel 4MA1 • November 2025 • Paper 2H • Q18 $\frac{19}{40}$

IDEA 02 • THREE-CIRCLE VENN DIAGRAMS

- Q9** Dr Eslam • Middle first $10 - 4 = 6$
- Q10** Practice • Easy $M = 16$, not carrots $3 + 6 = 9$: $\frac{9}{16}$
- Q11** Practice • Hard regions: all 19, TC only 16, CM only 2, TM only 5, C only 3, M only 1, T only 4 (a) $\frac{44}{50} = \frac{22}{25}$
(b) $\frac{21}{44}$
- Q12** Practice • Hard (a) 11 in A only, 6 in $B \cap C$ only, 1, 4, 10 outside
(b) $\{2, 5, 7, 8\}$ (c) $\{9, 12\}$ (d) 6 (e) $\frac{2}{13}$
- Q13** Practice • Hard (a) all 5; DL only 4, DT only 6, LT only 2; D only 4, L only 6, T only 3; outside 2 (b) $\frac{5}{9}$
- Q14** Edexcel 4MA1 • November 2021
• Paper 1H • Q16 $x = 7$, and $\frac{14}{25}$
- Q15** Edexcel 4MA1 • June 2022 • Paper 2H • Q16 36, 44, 35, $\frac{18}{53}$
- Q16** Edexcel 4MA1 • June 2024 • Paper 2H • Q16 $\frac{17}{28}$, 11, 28
- Q17** Edexcel 4MA1 • June 2025 • Paper 2H • Q20 (b) 16, (c) $\frac{18}{43}$

IDEA 03 • TREE DIAGRAMS: INDEPENDENT EVENTS

- Q18** Dr Eslam • Multiply along $0.09; 2(0.3)(0.7) = 0.42$
- Q19** Practice • Easy $1 - \frac{2}{3} \times \frac{1}{4} = \frac{5}{6}$
- Q20** Practice • Medium Monday $0.55 + 0.35 \neq 1$ (should be 0.45); Tuesday values swapped (0.25 rain, 0.75 not); no Tuesday branches after “does not rain”
- Q21** Practice • Hard (a) 0.98; 0.95 on each lower branch (b) $1 - 0.98 \times 0.95 = 0.069$
- Q22** Practice • Very Hard (a) C: $\frac{6}{14}, \frac{8}{14}$; D: $\frac{3}{10}, \frac{7}{10}$ (b) $\frac{2}{5}$ (c) $\frac{2}{5} \times \frac{7}{13} \times \frac{6}{9} = \frac{28}{195}$
- Q23** Edexcel 4MA1 • January 2020 • Paper 2HR • Q16 $\frac{7}{15}$, and 25
- Q24** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q15 $\frac{17}{25}$
- Q25** Edexcel 4MA1 • January 2021 • Paper 1H • Q14 0.54, 0.16
- Q26** Edexcel 4MA1 • January 2022 • Paper 1H • Q14 $\frac{14}{45}$
- Q27** Edexcel 4MA1 • January 2022 • Paper 1HR • Q13 (b) $\frac{25}{144}$ (c) 9 blue counters
- Q28** Edexcel 4MA1 • June 2023 • Paper 2H • Q13 $\frac{5}{9}$
- Q29** Edexcel 4MA1 • November 2023 • Paper 2H • Q11 $\frac{89}{160}$
- Q30** Edexcel 4MA1 • May 2024 • Paper 1H • Q13 $\frac{43}{90}$
- Q31** Edexcel 4MA1 • June 2025 • Paper 1H • Q14 0.18
- Q32** Edexcel 4MA1 • June 2025 • Paper 2HR • Q14 $\frac{7}{18}, \frac{386}{495}$
- Q33** Edexcel 4MA1 • November 2025 • Paper 1H • Q13 $\frac{17}{20}$

IDEA 04 • TREE DIAGRAMS: WITHOUT REPLACEMENT AND CONDITIONAL BRANCHES

- Q34** Dr Eslam • No replacement $\frac{6}{10} \times \frac{5}{9} = \frac{1}{3}$
- Q35** Practice • Medium (a) rain 0.2, 0.7/0.3; no rain 0.8, 0.05/0.95 (b) $0.8 \times 0.05 = 0.04$
- Q36** Practice • Hard (a) $\frac{6}{9}, \frac{3}{9}$ after red; $\frac{7}{9}, \frac{2}{9}$ after yellow (b) $1 - \frac{7}{10} \times \frac{6}{9} = \frac{8}{15}$
- Q37** Practice • Very Hard (a) A: 0.4, 0.6; after stop: 0.3, 0.7; after not stop: 0.8, 0.2 (b) A only 0.28, B only 0.48: lights B
- Q38** Practice • Very Hard (a) after circle: $\frac{3}{8}, \frac{5}{8}$; after square: $\frac{4}{8}, \frac{4}{8}$; first square $\frac{5}{9}$ (b) $\frac{20}{72} + \frac{20}{72} = \frac{5}{9}$
- Q39** Edexcel 4MA1 • January 2020 • Paper 2H • Q14 $\frac{21}{190}$
- Q40** Edexcel 4MA1 • June 2022 • Paper 1H • Q12 0.24
- Q41** Edexcel 4MA1 • May 2023 • Paper 1HR • Q13 0.63
- Q42** Edexcel 4MA1 • November 2024 • Paper 1H • Q13 0.63

IDEA 05 • LONGER TREES: THREE STAGES AND MORE BRANCHES

- Q43** Dr Eslam • At least one $1 - 0.4^3 = 0.936$
- Q44** Practice • Hard $0.7 + 0.3 \times 0.9 = 0.97$
- Q45** Practice • Hard $p^2 = \frac{49}{121} \Rightarrow P(10c) = \frac{7}{11}; 2 \times \frac{7}{11} \times \frac{4}{11} = \frac{56}{121}$
- Q46** Practice • Very Hard (a) $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{1}{2}$ (b) $P(\text{win}) = \frac{1}{6} + \frac{1}{2} \times \frac{1}{2} = \frac{5}{12} < \frac{1}{2}$: more likely to lose
- Q47** Practice • Very Hard (a) each jump: qualify 0.8, not 0.2 (b) $0.8 + 0.2 \times 0.8 = 0.96$
- Q48** Edexcel 4MA1 • May 2021 • Paper 2H • Q13 $0.54, 0.274$
- Q49** Edexcel 4MA1 • November 2021 • Paper 1H • Q15 $\frac{28}{81}, \frac{208}{729}$